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Comprehensive evaluation of water quality status for entire stretch of Yamuna River, India

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Abstract This study represented the first comprehensive assessment of the physicochemical water quality status of the entire Yamuna River stretch in India. The upper zone had "excellent-to-good" water quality index (WQI) with mean 5-day biochemical oxygen demand (BOD_5) values of 2.1 and 2.4 mg/L during monsoon and non-monsoon, respectively. The middle region was described by "poor-to-marginal" WQI with average BOD_5 values of 13.1 mg/L (monsoon) and 32.3 mg/L (non-monsoon). The low WQI observations at the midstream region were due to the negative impact of two major drains, namely Najafgarh and Shahdara, that carry partially treated effluents from industrial units. Further, BOD_5 decreased to 1.9 mg/L (monsoon) and 1.8 mg/L (non-monsoon) in the lower zone, and the WQI values improved to "good" and "excellent". The dilution and depuration effects of the Chambal, Sindh, Betwa, and Ken Rivers recovered the environmental conditions in downstream stations. The oxygen sag curve complied with the water quality status along the river stretch. Based on the principal component analysis, the Yamuna River was strongly influenced by dissolved mineral salts originating from atmospheric deposition, weathering of soils and rocks, and application of deicing chemicals and landfills. Moreover, organic and nutrient substances and biological activities resulting from the discharge of sewage, and the utilization of fertilizers in agriculture, were the second contributors to pollution. The statistical techniques employed in this work could be beneficial for decision-makers (government and stakeholders) to identify the pollution sources/factors and to determine the viability of water bodies for domestic applications.

Keywords Indian water standards · Monsoon and non-monsoon · Multivariate analysis · Water quality index · Yamuna River

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Introduction

India has 14 major rivers that contribute to 85% of the total surface flow (Mutiyar et al. 2018), and they signify vital environmental, historical, social, and economic aspects (Kumar et al. 2017). India also contains 44 normal and 55 small rain-fed rivers emanating from mountain regions at the coastal zones (Dash 2009).

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The rivers have been used for several agricultural, domestic, and industrial applications (Parmar and Bhardwaj 2015). Recently, most of the rivers have been subjected to heavy pollution loads caused by the discharge of untreated or partially treated wastewater and industrial effluents (Sharma et al. 2017a, b). The rivers have also been influenced by surface runoff from farmlands that carries anthropogenic chemicals, pesticides, and fertilizers (Bhardwaj et al. 2017). Accordingly, the river water becomes inappropriate for human consumption, even after the implementation of strict national and international regulations (Sharma et al. 2017a, b). Hence, comprehensive studies should be conducted to identify the pollution sources and evaluate the water quality in rivers.

The Central Pollution Control Board (CPCB) of India is responsible for reducing the water pollution in rivers via the actions of National Water Quality Monitoring Programme (CPCB 1980-1981; CPCB 1982-1983; CPCB 1999-2000; CPCB 2003, 2006). The CPCB of India aims at evaluating the water-related environmental issues and restoring and maintaining the wholesomeness of aquatic resources in water streams. The Government of India classified the country water bodies according to their quality and pollution status and then established plans to designate the best utilization of each class. For example, a Yamuna Action Plan was launched in 1993 by the Ministry of Environment and Forests, Government of India, to protect the water quality along the river, especially at the Delhi territory (Sharma and Singh 2009). Moreover, the Bureau of Indian Standards (BIS) has established the permissible and desirable limits of several water quality parameters for ascertaining the suitability of water for drinking purposes (IS 1991). Additionally, as per IS: 2296 (IS 1982), several categorizations including classes A, B, C, D, and E have been established to identify various applications of water resources. These standards have been used to describe the sampling procedures and assess the water quality conditions along the Indian rivers.

Yamuna River is one of the major water streams of the northern plains of India, and it is responsible for over 90% of irrigation practices (CPCB 1980-1981). The average annual rainfall along the river stretch varies from 65 to 125 cm, and the river has annual flow and usage of 10000 and 4400 m^3/s, respectively (Parmar and Keshari 2012). The Yamuna River is affected by different climatic conditions that vary between summer (45–50 °C) and winter (3–10 °C). The Yamuna River originates from a glacier called Yamunotri at an elevation of 6387 m on the southwestern slopes of Banderpoonch peaks of the Himalayan in the state of Uttarakhand (CPCB 1999-2000). It travels a total length of 1376 km by crossing seven states of India including National Capital Territory (NCT) of Delhi, and then joins the Ganges River at Allahabad (Sharma et al. 2017a, b). It accounts for more than 70% of water supplies in Delhi, and about 57 million inhabitants depend on the Yamuna River for their daily usage (Dash 2009). However, similar to other rivers, the Yamuna River illegally receives direct or indirect various pollutants from both point and non-point sources (CPCB 1999-2000). It delivers heavy loads of organics and nutrients, viz., approximately 80% from domestic sources (Sharma and Kansal 2011). Moreover, the water quality of the Yamuna River is negatively impacted by different types of anthropogenic inorganic and organic toxicants, as well as dissolved and particulate chemicals (CPCB 2003). Consequently, the river becomes incompatible for attaining its designated ecological functions (Dash 2009). In this context, comprehensive monitoring and assessment practices should be performed to evaluate the pollution status of the Yamuna River.

Previous studies have dealt with specific regions on the Yamuna River that are influenced by discharges from respective cities or states (Paliwal et al. 2007; Sharma and Singh 2009; Sharma and Kansal 2011; Parmar and Keshari 2012; Mutiyar et al. 2018). To the best of the authors' knowledge, this is the first reconnaissance study representing a detailed assessment of the pollution status of the whole stretch of the Yamuna River. The spatial variation of water quality parameters along the river including twelve monitoring stations, viz., from Paonta (upstream) to Pratappur (downstream), was investigated. The considered water quality parameters were temperature, pH, dissolved oxygen (DO), 5-day biochemical oxygen demand (BOD_5), total dissolved solids (TDS), electrical conductivity (EC), total alkalinity (TA), total hardness (TH), calcium-hardness (Ca-H), magnesium-hardness (Mg-H), ammonium (NH_4^+), nitrite (NO_2^-), nitrate (NO_3^-), phosphate (PO_4^{3-}), sulfate (SO_4^{2-}), chloride (Cl^-), fluoride (F^-), iron (Fe), and boron (B). The concentrations and levels of these parameters were compared with national and international regulations. The water quality index (WQI) of the river at each monitoring station was also demonstrated. The survey was conducted for two years

considering monsoon and non-monsoon seasons. The water quality assessment was explored regarding various statistical approaches such as PCA, cluster analysis, and correlation matrix.

Materials and methods

Study area

The Yamuna River is the fifth longest river in India having a total length of 1376 km and a drainage area of 366,223 km^2 in the Gangetic plain (Fig. 1). The origin of the Yamuna River is a frozen Champasar lake, which is situated in Bandarpoonch Glacier (38° 59′ N, 78° 27′ E) in Har-Ki-Doon mountain region of Uttarakhand (CPCB 1999-2000). The Yamuna River passes through several snowcapped mountains and forested valleys in Himalaya before reaching the Indo-Gangetic plains of north India. Finally, the river joins the River Ganga at Allahabad, Uttar Pradesh, state of India.

Sampling sites and sample collection

The monitoring stations used to cover the entire Yamuna stretch were Paonta, Kalanaur, Mawi, Palla, Delhi, Mohana, Mathura, Agra, Etawah, Auraiya, Hamirpur, and Pratappur. The water quality stations along with their latitude and longitude data and the catchment areas are provided in the supplementary file (Table S1). Grab samples of water were collected from the center of the river at about 0.3–0.5 m beneath the water surface. The sampling protocol was performed during two different seasons of the year (2014–2015), i.e., monsoon (July to October) and non-monsoon (November to June). The samples were stored in sampling kits, preserved at 4 °C, and brought to the laboratory within 2 h for analysis.

Analytical analysis

The water quality parameters were analyzed as per the standard methods of the American Public Health Association (APHA 2005). The parameters of temperature, pH, DO, and EC were determined in situ using a portable water analysis kit (YSI Professional Plus handheld multi-parameter meter, USA). Trace metals were analyzed using acid digestion, following standard methods (APHA 1998). In brief, an aliquot of 100 mL was digested with nitric acid and perchloric acid at the ratio of 4:1 (*v*/*v*). The digested solution was cooled, diluted with deionized water, and filtered through Whatman no. 42 filter paper. Further, the concentrations of elemental compounds were determined by an atomic absorption spectrophotometer (ECIL 4141, India).

Statistical analysis

Pearson's correlation coefficient, also known as "*r*," was estimated to determine the relationships between the water quality parameters. The correlation coefficient was statistically significant at $p < 0.05$, based on the sample size. The correlation is considered "significant decrease" or "significant increase" at "*r*" values near − 1 or 1, respectively (Gholizadeh et al. 2016).

Principal component analysis (PCA) is a multivariate statistical tool employed to recognize the leading causes of variance in a collection of data that contains a large number of interrelated variables (Nasr and Zahran 2016). PCA rotates a set of observations into orthogonally correlated variables called PCs, where the new axes represent the directions of maximum variances. This step is used for data reduction while keeping as much information about the original data as possible (Jankowska et al. 2017). The PC having an eigenvalue higher than unity retains most of the variability of the dataset and, hence, considered for evaluation. PCs can be expressed by Eq. 1 (Gholizadeh et al. 2016).

$$Z_{ij} = \alpha_{i1}X_{1j} + \alpha_{i2}X_{2j} + \alpha_{i3}X_{3j} + \ldots + \alpha_{im}X_{mj} \quad (1)$$

where Z is the component score; α is the component loading; X is the measured variable; and i, j, and m are the numbers of the component, sample, and variables, respectively.

Loadings, which range between − 1 and 1, are used to explain the contribution of the measured parameter in a particular PC. A loading value of almost zero indicates that the variable is inadequately correlated with the component, whereas that near − 1 or 1 implies that the variable strongly influences the component (Nasr and Zahran 2016). Additionally, a variable having a positive loading sign shows a direct effect on the factor, and it acts oppositely to other variables having negative loadings.

Cluster analysis was carried out to determine the similarity and dissimilarity in the spatial variation

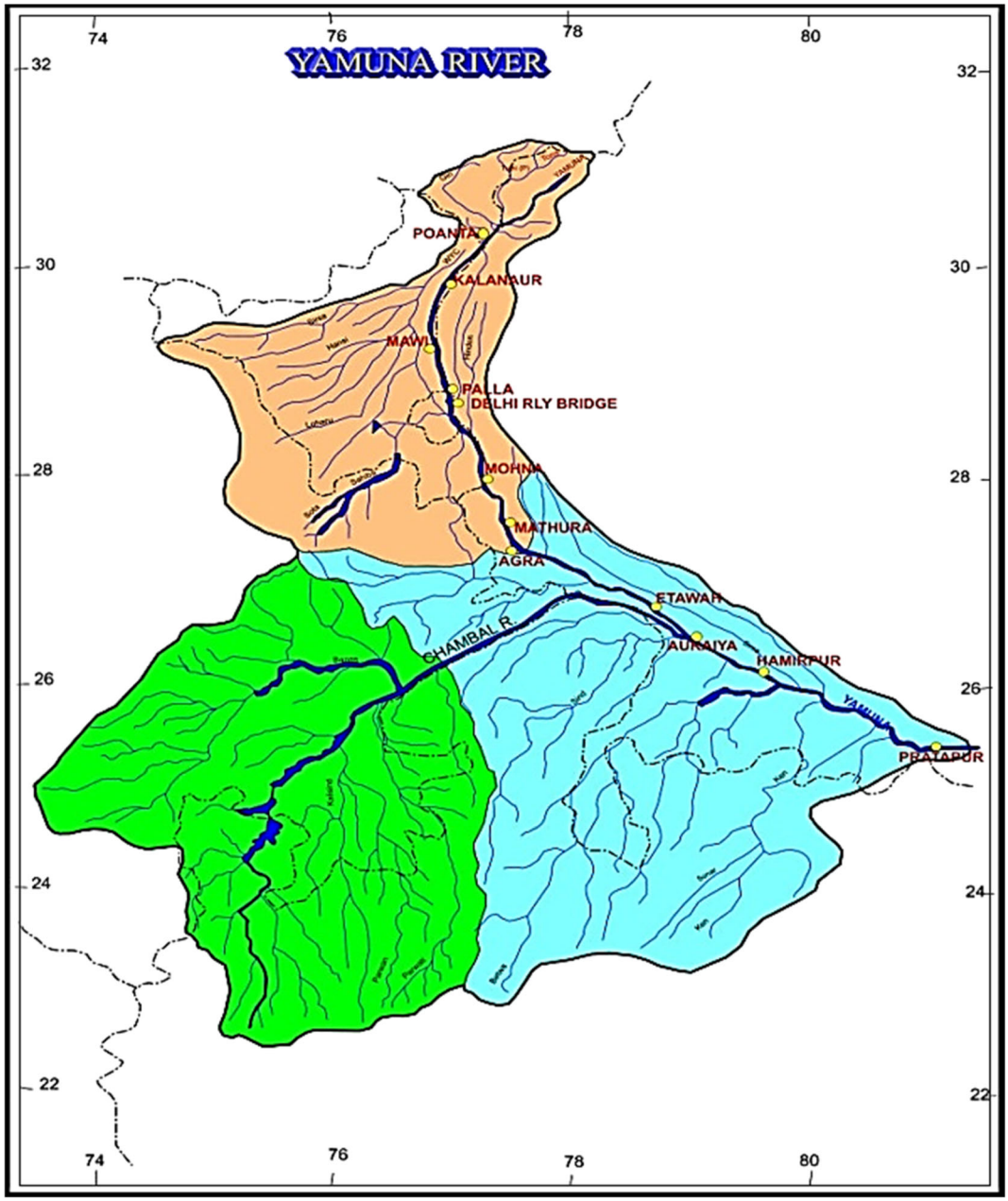

Fig. 1 Layout of the Yamuna River under investigation

among sampling sites. Cluster analysis is accomplished when the assembled observations have high internal homogeneity within a cluster and high external heterogeneity among groups (Singh et al. 2004). Cluster analysis was performed on the normalized data using Ward's method and squared Euclidean distances as a measure of similarity (Willet 1987). All calculations of statistical analysis were performed using MATLAB software (MathWorks 2017).

Results and discussion

Water quality parameters along the Yamuna River

The concentrations of water quality parameters at the monitoring stations during monsoon and non-monsoon were estimated (Supplementary file; Tables S2 and S3). These values were compared with the allowable limits listed in Table 1, as per Indian Standards (IS 1982,

1991). The Indian Standards have identified the usage of water for different purposes according to classes A, B, C, D, and E (Supplementary file; Table S4).

Temperature

The average temperature data were 33.2 ± 2.2 °C and 27.7 ± 2.0 °C during the monsoon and non-monsoon seasons, respectively. The temperature observations showed insignificant ($p > 0.05$) spatial variation among the monitoring stations. This result could be due to the narrow range of latitude along the Yamuna River (i.e., between 25° 21′ 17″ and 30° 25′ 31″). The temperature of air affects the river temperature, which consequently impacts the water quality condition. No permissible limit of temperature has been identified by the Indian Standards of IS: 2296 (IS 1982) and IS: 10500 (IS 1991).

pH

The pH value, which identifies the acidic or alkaline condition of aqueous solutions, is considered an important parameter for the assessment of water quality in rivers (APHA 2005). The pH levels in water bodies are influenced by the chemical composition of sediments and metal complexes including SO_x and NO_x compounds. During the monsoon season, the minimum and maximum pH values were reported at Agra (pH 7.62) and Kalanaur (pH 8.07), respectively. In the non-monsoon period, the pH values varied between 7.60 at Delhi and 8.47 at Etawah. These results suggested that the pH observations were acceptable (pH 6.5–8.5) according to IS: 10500 (see Table 1). Similar regulations have been reported by the World Health Organization (WHO) with an allowable pH range of 6.5–8.5 (WHO 2011). Additionally, the pH values indicated that the Yamuna River could be used for drinking purposes after disinfection, as specified in class A (IS 1982).

Dissolved oxygen

Dissolved oxygen (DO) is a vital element for the growth, production, and survival of aquatic life, and its deficiency is an indicator of sewage pollution (CPCB 1982-1983). The DO concentrations ranged as 4.5–7.9 mg/L during monsoon and 1.7–6.6 mg/L in non-monsoon. Additionally, the average value of DO during the monsoon period (5.9 mg/L) was higher than that during non-monsoon (4.7 mg/L). This result could be due to the wave action, sufficient turbulence, and high oxygen diffusion from the surrounding atmosphere during monsoon (Sharma and Kansal 2011). The DO observations were within the permissible limit of class B (DO > 5 mg/L) at Paonta, Kalanaur, Mawi, and Palla (IS 1982). These sites receive water from drains containing low organic pollution, and thus, they could be used for some applications such as outdoor bathing. However, the DO levels declined for the sites of Delhi, Mohana, and Agra and reached class C (DO > 4 mg/L) (IS 1982). This result could be due to the photosynthetic activity of aquatic life as well as aerobic decomposition of the organic matter by microorganisms (CPCB 2006). Further, the DO profile increased again with the flow of the river from Etawah to Pratappur.

Five-day biochemical oxygen demand

BOD_5 is the major criterion used for stream pollution monitoring and control (APHA 2005). BOD_5 has broad applications in environmental studies, and it provides sufficient information on the biologically oxidizable organic matter (Dash 2009). During monsoon, the average BOD_5 concentration was 6.6 mg/L, which ranged from 0.9 mg/L at Pratappur to 26.8 mg/L at Agra. In non-monsoon, BOD_5 ranged from 1.5 mg/L at Paonta to 53.9 mg/L at Delhi with an average value of 14.7 mg/L. The average BOD_5 level in the Yamuna River from Paonta to Palla was within the criterion of class B (≤ 3 mg/L). The low BOD_5 levels at these locations could be due to the absence of significant contribution of wastewater from the surrounding catchment. However, the significant increase in BOD_5 was noticed downstream of Palla, which could be due to receiving a discharge from Najafgarh Drain, i.e., the biggest drain that carries about 40% of wastewater from Delhi (Parmar and Bhardwaj 2015). Additionally, the Shahdara Drain carries about 30% of wastewater from Delhi and then connects to the Yamuna River near Kalindi Kunj in Delhi. The BOD_5 values in these drains normally exceed 100 mg/L (CPCB 2006). The increase in BOD_5 levels at Mathura could be due to the contribution of the organic load from two sewage treatment plants (oxidation pond type) having a combined capacity of 28 MLD. The increase in BOD_5 levels at Agra could also be due to the industrial activities and

Table 1 Indian Standards of the studied water quality parameters along the Yamuna River

Water quality parameter	IS: 2296 (1982)					IS: 10500 (1991)	
	Class					Desirable limit	Permissible limit
	A	B	C	D	E		
pH	6.5–8.5	6.5–8.5	6.0–9.0	6.5–8.5	6.0–8.5	6.5–8.5	No relaxation
DO (mg/L)	≥ 6	≥ 5	≥ 4	≥ 4	–	–	–
BOD (mg/L)	≤ 2	≤ 3	≤ 3	–	–	–	–
TDS (mg/L)	≤ 500	–	≤ 1500	–	≤ 2100	500	2000
EC (µƱ/cm)	–	–	–	≤ 1000	≤ 2250	–	–
TA (mg/L)	–	–	–	–	–	200	600
TH (mg/L)	≤ 300	–	–	–	–	300	600
Ca-H (mg/L)	≤ 200	–	–	–	–	75	200
Mg-H (mg/L)	≤ 100	–	–	–	–	30	100
NH_4^+ (mg/L)	–	–	–	≤ 1.2	–	0.5	No relaxation
NO_3^- (mg/L)	≤ 20	–	≤ 50	–	–	45	100
SO_4^{2-} (mg/L)	≤ 400	–	≤ 400	–	≤ 1000	200	400
Cl^- (mg/L)	≤ 250	–	≤ 600	–	≤ 600	250	1000
F^- (mg/L)	≤ 1.5	≤ 1.5	≤ 1.5	–	–	1.0	1.5
Fe (mg/L)	≤ 0.3	–	≤ 0.5	–	–	0.3	1.0
B (mg/L)	–	–	–	–	≤ 2.0	1.0	5.0

DO dissolved oxygen, *BOD* biochemical oxygen demand, *TDS* total dissolved solids, *EC* electrical conductivity, *TA* total alkalinity (as $CaCO_3$), *TH* total hardness (as $CaCO_3$), *Ca-H* calcium hardness (as $CaCO_3$), *Mg-H* magnesium hardness (as $CaCO_3$), NH_4^+ ammonium, NO_3^- nitrate, SO_4^{2-} sulfate, Cl^- chloride, F^- fluoride, *Fe* iron, and *B* boron

discharges from Buria Ka Nagla sewage treatment plants. Further, there are no major towns located along a distance of 170 km between Agra and Etawah, but the water river increases due to the addition of subsoil water (CPCB 2003). Hence, the BOD_5 values significantly declined at Etawah due to the natural bacteriological/biological oxidation of organic matter. The BOD_5 levels continued to decrease after Etawah and varied between 1 and 3 mg/L for the sites of Auraiya, Hamirpur, and Pratappur. This trend could be because of the dilution effects from some tributaries such as Chambal, Sindh, Betwa, Rind, and Ken. Additionally, no major industrial settlements exist in the catchment between Etawah and Pratappur (CPCB 1980-1981).

Total dissolved solids

The TDS levels ranged between 117 and 353 mg/L during the monsoon period and between 169 and 940 mg/L during the non-monsoon season. The minimum and maximum TDS values were reported at Paonta and Agra, respectively (Supplementary file; Tables S2 and S3). During monsoon, the Yamuna River receives significant amounts of precipitation, which may be beyond its conveyance capacity. This trend endorses the dilution effect and decreases the TDS concentration in the river (Singh et al. 2004). On the contrary, the river flow experiences water reduction during the non-monsoon period, and some short rivers become dry. Hence, non-monsoon recorded higher TDS values than monsoon. During the monsoon season, the TDS values suggested that the river could be suitable for drinking purposes after disinfection (TDS ≤ 500 mg/L; class A) (IS 1982). However, the water at some sites such as Delhi, Agra, and Auraiya, which contained TDS over 500 mg/L, could be appropriate for drinking after conventional treatment. All sites observed acceptable TDS levels ≤ 1500 mg/L for class C and ≤ 2100 mg/L for class E (IS 1982). According to the TDS data, the Yamuna River was suitable for drinking purposes after conventional treatment and disinfection, and it could be used for irrigation and industrial applications.

Electrical conductivity

EC is used to describe the cations and anions in water bodies, and it is directly correlated with the concentration of dissolved ions in solutions (APHA 1998). The minimum EC values of 195 and 281 μ℧/cm were reported at Paonta during the monsoon and non-monsoon seasons, respectively. Additionally, the highest EC observations were found at Agra with values of 588 μ℧/cm during monsoon and 1567 μ℧/cm during non-monsoon. These results indicated that water at Delhi, Mohana, Mathura, Agra, Etawah, Auraiya, and Hamirpur exceeded the allowable EC of 1000 μ℧/cm according to class D during non-monsoon (IS 1982). This category is adequate for the propagation of wildlife and fish culture (Supplementary file; Table S4). Additionally, all sites that contained permissible limits of EC according to class E (≤ 2250 μ℧/cm; Table 1) were applicable for industrial cooling and irrigation (IS 1982). Moreover, only Agra slightly exceeded the maximum permissible limits of 1500 μ℧/cm, as prescribed by WHO (2011).

Total alkalinity and total hardness

According to the Indian Standards (IS 1991), the desirable and permissible levels of alkalinity in water bodies are 200 and 600 mg/L, respectively. The river water at several sites such as Delhi, Mohana, Mathura, Agra, Etawah, Auraiya, and Hamirpur exceeded the desirable concentration of alkalinity during the non-monsoon season. High-alkalinity water may be unsuitable for plant irrigation (Ayers and Westcot 1994).

During monsoon, the average TH of the Yamuna River was 140 ± 30 mg/L, which ranged between 90 and 181 mg/L. During non-monsoon, the minimum and maximum TH values recorded 130 and 304 mg/L, respectively. These TH observations were mainly linked to the components of dissolved calcium and magnesium salts. The TH value at Agra during non-monsoon exceeded the allowable level of 300 mg/L for class A (IS 1982). However, most sites contained the permissible limit of TH (≤ 300 mg/L) that could be applied for drinking purposes.

Nutrient (nitrogen and phosphorus)

The NH_4^+ concentration varied from 0.07–0.08 mg/L at Paonta to 5.78–6.64 mg/L at Delhi. The NH_4^+ levels at Delhi and Agra exceeded those of IS: 2296 (class D ≤ 1.2 mg/L) and IS: 2296 (0.5 mg/L), suggesting the contribution of industrial sources, agricultural fertilizers, and livestock facilities. NO_2^- and NO_3^- occur naturally in surface water as a part of the nitrogen cycle. The allowable limits of NO_2^- in water bodies is not reported in either IS: 2296 (IS 1982) or IS: 10500 (IS 1991). This observation could be because NO_2^- is not a stable nitrogen element, as it can occur due to the oxidation of ammonia (nitrification) or the reduction of nitrate. The NO_3^- concentrations ranged as 0.30–17.38 mg/L in monsoon and 0.96–17.89 mg/L in non-monsoon. The least and maximum NO_3^- concentrations were recorded at Pratappur and Mohana, respectively. The NO_3^- concentrations along the Yamuna River obeyed the allowable limits according to IS: 2296 (IS 1982) and IS: 10500 (IS 1991). However, a high concentration of NO_3^- at some locations such as Delhi, Mohana, and Mathura indicated anthropogenic contamination, runoff from agricultural land, and mixing of human and animal wastes (Djemai et al. 2017). The enrichment of surface water with nitrogen components can result in rapid growth of aquatic plants leading to oxygen depletion. Similarly, the high concentrations of PO_4^{3-} at Delhi, Mohana, and Mathura designated the impact of drainage discharge. The decline in nutrient concentrations during the monsoon period could be due to the high dilution capacity and aeration properties (Sharma et al. 2017a, b).

Sulfate

SO_4^{2-} can enter water streams due to sewage discharge and surface runoff from the contaminated regions. The SO_4^{2-} concentration along the Yamuna River varied from 17 to 51 mg/L with a mean concentration of 34 mg/L in monsoon, as well as 18–111 mg/L with a mean of 55 mg/L in non-monsoon. The lowest SO_4^{2-} concentration was observed at Kalanaur and the highest at Agra during both seasons. These observations revealed that the concentrations of SO_4^{2-} at the sampling sites were below the maximum desirable limit of 200 mg/L, as per the Indian Standards (IS 1991). In addition, the SO_4^{2-} levels complied with class A of IS: 2296 (≤ 400 mg/L; Table 1).

Chloride

The average Cl^- concentrations along the Yamuna River were 40.5 ± 16.6 and 112.4 ± 76.5 mg/L during the monsoon and non-monsoon seasons, respectively. The Cl^- content in water streams depends on the pollution load, in which a high value of chloride indicates a contribution of sewage contamination (CPCB 1982-1983). Cl^- is highly mobile, and it tends to transfer from the soil-water interface to plant roots and then accumulates in the leaves. The highest Cl^- concentration of 226.7 mg/L was reported at Agra during the non-monsoon period. It was found that all sites during both seasons contained allowable Cl^- concentrations according to class A (≤ 250 mg/L) (IS 1982) and they followed the desirable limit of IS: 10500 (Table 1). This finding indicated the absence of Cl^- toxicity along the longitudinal profile of the Yamuna River.

Fluoride

Drinking water containing F^- with a permissible limit of 0.5–1.5 mg/L is beneficial to human health (WHO 2011). At this range, F^- can prevent tooth decay by enhancing the remineralization of dental enamel. However, the excess intake of F^- may result in multiple health implications such as dental and skeletal fluorosis, and osteosclerosis, and it can severely affect the physiology of various vital organs such as the liver, kidney, and heart (Ayers and Westcot 1994). The mean F^- concentration in the Yamuna River was 0.35 mg/L in monsoon, and it increased to 0.44 mg/L in the non-monsoon period. At all sampling sites, the concentrations of F^- were allowable according to the WHO desirable limit (WHO 2011), and the water could be utilized for drinking after a disinfection step.

Iron

Fe is an abundant metallic element in the earth's crust, and it occurs naturally in groundwater and surface water giving the metallic taste to drinking water (APHA 2005). Fe can enter the river systems from anthropogenic sources such as untreated or partially treated industrial effluents, and metal chelates from various industries (Mutiyar et al. 2018). The concentrations of Fe ranged as 0.17–0.90 mg/L during monsoon and 0.07–0.81 mg/L during non-monsoon. Several locations such as Delhi, Mohana, Mathura, and Agra exceeded the acceptable limit of Fe in drinking water (0.3 mg/L) (IS 1991). The high concentration of Fe at these locations could be attributed to the discharge of Shahdara Drain that carries effluents from the electroplating and steel-processing industries situated in Badli, Jhilmil, and Wazirpur zones. Additionally, these areas were influenced by atmospheric precipitation and excess application of chemical fertilizers at surrounding agricultural fields. Fe is a beneficial element in human nutrition, and its deficiency can cause different types of anemia, Cooley's anemia, and thalassemia. However, Fe overload may damage the body tissues and cause hemochromatosis, hemosiderosis, and lesions of the gastrointestinal tract (Gupta et al. 2018). Additionally, high Fe contents in drinking water can lead to serious health problems including rapid and shallow respiration, cardiac arrest, and diabetes (WHO 2011).

Boron

B occurs naturally in groundwater as an earth element, whereas its presence in surface water is a consequence of the discharge of wastewater and sewage effluents (CPCB 2006). B is an essential element for plant growth, but it becomes harmful to certain vegetation and crops when exceeding 0.2 mg/L in irrigation water (Ayers and Westcot 1994). Along the Yamuna River, B ranged as 0.02–0.22 mg/L with an average value of 0.11 mg/L in monsoon and 0.02–0.40 mg/L with an average of 0.22 mg/L in non-monsoon. These values were allowable according to IS: 2296 (≤ 2.0 mg/L for class E) and IS: 10500 (1.0 mg/L). The minimum concentrations of B were found at Kalanaur during monsoon and Paonta and Mawi during non-monsoon. However, the highest concentrations of B were detected at Etawah and Mathura during monsoon and non-monsoon, respectively. High B concentrations could be linked to geothermal activities and water-rock reactions (Kumar et al. 2017).

Water quality index

WQI was estimated for the twelve sampling stations along the Yamuna River following the protocol of the Canadian Council of Ministers of the Environment (CCME) (CCME 2001). The WQI classifications are "excellent" (95–100), "good" (80–94), "fair" (65–79), "marginal" (45–64), and "poor" (0–44) (Supplementary file; Table S5). Figure 2 shows the longitudinal profile

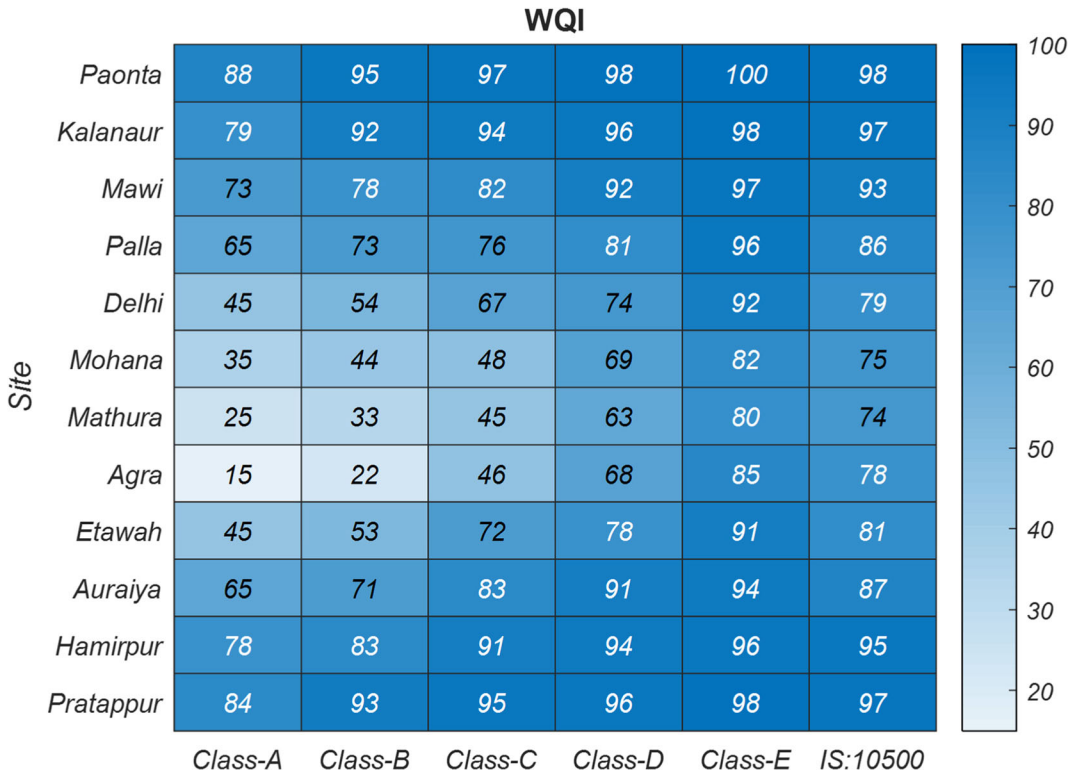

Fig. 2 Water quality index (WQI) at each monitoring station along the Yamuna River, according to class A to E, and IS: 10500

of WQI estimated at each class according to the Indian Standards (IS 1982, 1991). The upper zones of the Yamuna River from Paonta to Palla had "excellent-to-good" WQI, whereas the middle locations from Delhi to Etawha were described by "poor-to-marginal" WQI. The decrease in WQI at the midstream of Yamuna could be due to the growing density of the human population on the river banks along with poor sanitation practices by inhabitants (Sharma et al. 2017a, b). Further, the WQI values in the lower zone from Auraiya to Pratappur were enhanced and, thus, recorded "good-to-excellent" classifications. This result could be due to the dilution effect from the Chambal, Betwa, and Ken Rivers (CPCB 1999-2000). The WQI observations at Pratappur for all classes were similar to those at Paonta indicating that the water characteristics at the downstream region were recovered.

The water quality at Paonta was "good" for drinking purposes after disinfection and "excellent" for other applications such as outdoor bathing and drinking after conventional treatment (IS 1982). The Yamuna River at Kalanaur and Mawi had "good" capabilities for drinking purposes after treatment and chlorination, while it recorded "excellent" applications for irrigation and industrial cooling systems. The range of WQI for the Yamuna River at Palla dropped below "fair" category in classes A, B, and C, which could be linked to the high pollution load in the river and limitation in the treatment facilities. The water quality at Delhi, Mohana, Mathura, and Agra were "poor" for most applications except for industrial cooling, and controlled waste disposal. The "poor" category of WQI was predominant due to the low DO levels as a result of drainage wastewater discharge and anthropogenic activities. Mohana is located about 50 km downstream of Delhi, and no rivers or rivulets join the Yamuna between these two cities. The poor water quality at Mathura could be attributed to the presence of Mathura refinery, which is a large industrial unit in the catchment and its effluent joins the Yamuna River at about 40 km downstream. Moreover, water quality at Auariya and Hamirpur were "excellent" for the propagation of wildlife and fisheries (IS 1982). Additionally, the river at Pratappur can be used for drinking purposes after disinfection (class A) and outdoor bathing (class B) throughout the year.

The main factor affecting WQI of class A was BOD_5, in which a low value of $BOD_5 < 6.6$ mg/L resulted in "fair-to-good" WQI, whereas, the "poor" WQI was

observed at $BOD_5 \geq 6.6$ mg/L. High BOD_5 levels in the river can deteriorate aquatic organisms and increase eutrophication (Kumar et al. 2017). Similarly, BOD_5 provided the major impact on the river according to classes B and D. At $BOD_5 < 2.7$ mg/L, class B and class D became "good" and "excellent," respectively. However, a "poor" classification was observed for both classes B and D at $BOD_5 \geq 2.7$ mg/L. Class C was highly correlated with the DO concentrations in the river, being "marginal" WQI at DO < 4.9 mg/L, and "good" WQI at DO ≥ 4.9 mg/L. The concentrations of DO in water streams depend on multiple factors such as surface agitation, photosynthetic activity, temperature, and pollution load. Chloride was the main factor affecting class E, in which $Cl^- \geq 55.2$ mg/L and < 55.2 mg/L exhibited "good" and "excellent" WQIs, respectively.

Correlation matrix

A correlation matrix was used to identify relationships among the 15 water quality parameters of the Yamuna River obtained during monsoon (Table 2) and non-monsoon (Table 3).

Correlation matrix for monsoon

A significant and inverse correlation was observed between DO and BOD_5 (r −0.721, $p < 0.05$), which could be due to the consumption of DO by aerobic species during the assimilation of organics in polluted water (Singh et al. 2017). Additionally, DO was oppositely correlated with NO_3^- (r −0.666, $p < 0.05$), suggesting the interaction between aerobic and nitrification processes. EC was significantly ($p < 0.05$) correlated with TDS (r 1.000), TA (r 0.826), TH (r 0.947), Ca-H (r 0.624), Mg-H (r 0.765), SO_4^{2-} (r 0.902), Cl^- (r 0.951), and F^- (r 0.682). This result suggested that the species of Ca-H, Mg-H, Cl^-, and SO_4^{2-} made the strongest contribution to conductivity. Furthermore, the high correlation among the ionic charges could be linked to the geochemical composition of the river basin. The significant correlation between NH_4^+ and PO_4^{3-} (r 0.746, $p < 0.05$) implied the discharge of nutrient species from common sources. The inverse correlation between temperature and DO could be because large amounts of oxygen tend to dissolve in cooler water.

Table 2 Correlation matrix among water qualities parameters of the Yamuna River during monsoon season. Italicized values represent significant correlations ($p < 0.05$)

	Temp.	pH	DO	BOD	TDS	EC	TA	TH	Ca-H	Mg-H	NH_4^+	NO_2^-	NO_3^-	PO_4^{3-}	SO_4^{2-}	Cl^-	F^-	Fe	B
Temp.	1.0																		
pH	−0.3	1.0																	
DO	−0.3	0.0	1.0																
BOD	0.3	−0.4	*−0.7*	1.0															
TDS	0.3	*−0.7*	0.0	0.5	1.0														
EC	0.3	*−0.7*	0.0	0.5	*1.0*	1.0													
TA	0.1	*−0.7*	0.1	0.4	*0.8*	*0.8*	1.0												
TH	0.2	*−0.8*	0.0	0.5	*0.9*	*0.9*	*0.8*	1.0											
Ca-H	0.2	−0.5	−0.5	*0.7*	*0.6*	*0.6*	0.5	*0.6*	1.0										
Mg-H	0.3	*−0.7*	0.4	0.2	*0.8*	*0.8*	*0.7*	*0.8*	0.1	1.0									
NH_4^+	0.4	−0.3	−0.3	0.4	0.4	0.4	0.3	0.4	0.5	0.1	1.0								
NO_2^-	0.2	−0.5	*−0.6*	*0.8*	0.5	0.5	0.3	0.5	*0.8*	0.2	0.0	1.0							
NO_3^-	0.0	0.4	*−0.6*	0.3	0.0	0.0	−0.3	−0.2	0.4	−0.6	0.4	0.2	1.0						
PO_4^{3-}	0.2	0.0	−0.5	0.5	0.5	0.5	0.1	0.3	*0.7*	−0.1	*0.7*	0.3	*0.8*	1.0					
SO_4^{2-}	0.3	*−0.8*	−0.3	*0.6*	*0.9*	*0.9*	*0.8*	*0.9*	*0.8*	*0.6*	0.5	*0.6*	0.1	0.5	1.0				
Cl^-	0.2	−0.5	−0.1	0.6	*1.0*	*1.0*	*0.8*	*0.9*	*0.7*	*0.7*	0.5	0.5	0.1	*0.6*	*0.9*	1.0			
F^-	0.2	−0.4	0.0	0.4	*0.7*	*0.7*	0.3	*0.6*	0.3	0.5	0.0	0.5	0.1	0.4	*0.6*	*0.7*	1.0		
Fe	0.2	0.3	−0.3	−0.1	−0.3	−0.3	−0.5	−0.4	−0.3	−0.4	−0.2	−0.2	0.2	0.0	−0.4	−0.3	−0.2	1.0	
B	0.2	*−0.7*	0.5	0.1	*0.6*	*0.6*	*0.8*	*0.7*	0.1	*0.8*	−0.1	0.1	*−0.7*	−0.3	0.5	0.5	−0.2	0.2	1.0

Table 3 Correlation matrix among water qualities parameters of the Yamuna River during non-monsoon season. Italicized values represent significant correlations ($p < 0.05$)

	Temp.	pH	DO	BOD	TDS	EC	TA	TH	Ca-H	Mg-H	NH_4^+	NO_2^-	NO_3^-	PO_4^{3-}	SO_4^{2-}	Cl^-	F^-	Fe	B
Temp.	1.0																		
pH	0.3	1.0																	
DO	− 0.2	0.4	1.0																
BOD	0.3	− 0.5	*− 0.9*	1.0															
TDS	*0.7*	0.3	− 0.4	0.5	1.0														
EC	*0.7*	0.3	− 0.4	0.5	*1.0*	1.0													
TA	*0.6*	0.2	− 0.5	0.6	*1.0*	*1.0*	1.0												
TH	*0.6*	0.2	− 0.4	0.5	*1.0*	*1.0*	*1.0*	1.0											
Ca-H	0.5	− 0.1	*− 0.8*	*0.8*	*0.8*	*0.8*	*0.9*	*0.8*	1.0										
Mg-H	*0.6*	0.5	0.1	0.1	*0.9*	*0.9*	*0.8*	*0.9*	0.4	1.0									
NH_4^+	0.3	0.1	*− 0.7*	*0.6*	*0.7*	*0.7*	*0.7*	*0.7*	*0.8*	0.5	1.0								
NO_2^-	0.3	*0.6*	0.1	− 0.1	*0.6*	*0.6*	0.5	0.5	0.4	0.5	0.4	1.0							
NO_3^-	0.0	*− 0.7*	*− 0.8*	*0.8*	0.0	0.0	0.1	0.0	0.5	− 0.4	0.2	− 0.4	1.0						
PO_4^{3-}	0.4	− 0.3	*− 0.9*	*0.9*	0.5	0.5	0.5	0.5	*0.8*	0.0	0.5	0.0	*0.8*	1.0					
SO_4^{2-}	0.5	0.3	− 0.4	0.4	*0.9*	*0.9*	*0.9*	*0.9*	*0.8*	*0.7*	*0.7*	*0.7*	0.0	0.5	1.0				
Cl^-	*0.6*	0.2	− 0.4	0.5	*1.0*	*1.0*	*0.9*	*1.0*	*0.8*	*0.8*	*0.7*	0.5	0.0	0.5	*0.9*	1.0			
F^-	*0.7*	0.1	− 0.6	*0.7*	*0.9*	*0.9*	*0.9*	*0.9*	*0.9*	*0.6*	*0.7*	0.5	0.2	*0.7*	*0.8*	*0.9*	1.0		
Fe	0.3	− 0.6	*− 0.8*	*0.9*	0.3	0.3	0.4	0.2	0.5	− 0.1	0.5	− 0.3	*0.8*	*0.7*	0.2	0.3	0.4	1.0	
B	0.5	− 0.2	− 0.5	*0.7*	*0.7*	*0.7*	*0.7*	*0.6*	*0.8*	0.3	0.4	0.2	0.5	*0.7*	0.5	*0.6*	*0.8*	*0.7*	1.0

Correlation matrix for non-monsoon

EC, TA, TH, Ca-H, Mg-H, SO_4^{2-}, Cl^-, and F^- showed significant ($p < 0.05$) and positive correlations with TDS, demonstrating the release of ions from the minerals in soils and rocks over which the river flows. Moreover, the high correlation between Fe and B (r 0.668, $p < 0.05$) suggested pollution from the effluents of industrial units containing excessive iron and boron concentrations. Moreover, the water quality parameters associated with anthropogenic activities were positively correlated ($p < 0.05$). The negative correlations of NH_4^+–DO (r − 0.724, p 0.05) and PO_4^{3-}–DO (r − 0.931, p 0.05) demonstrated that the enrichment of nutrients in water streams is favorable for the growth of aquatic plants that consume a high amount of DO. The DO–BOD_5 correlation (r − 0.932, $p < 0.05$) suggested that the low DO levels at several locations were invariably accompanied by high BOD_5 values. The negative correlation between temperature and DO implied that an increase in temperature encourages the biological activity of aquatic organisms, leading to DO consumption.

Principal component analysis

Figure 3 a (monsoon) and b (non-monsoon) show the water quality variables at a z-scale transformation with a mean of zero and variance = one. The standardization step is essential before PCA when the variables have different units of measurements so that each parameter should receive equal weight in the analysis (Nasr and Zahran 2016). Table 4 lists the estimated PCs and their eigenvalues, as well as the cumulative percentage of variance retained by each PC. The first two PCs were observed with eigenvalues higher than 1, and they provided total cumulative variances of 70.6% and 83.7% for the monsoon and non-monsoon seasons, respectively. Hence, PC1 and PC2 were used in this study for the identification of the most important variables that could describe the spatial variations along the Yamuna River (Fig. 4).

PCs for monsoon

PC1 accounted for 48.6% of the total variance, and it showed explanatory variables of pH, TDS,

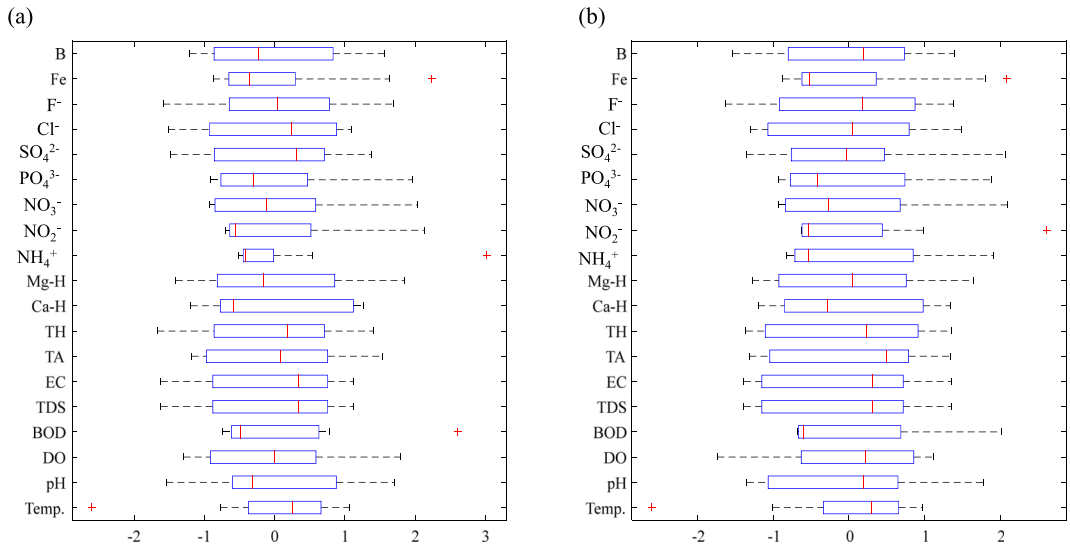

Fig. 3 Standardization of physicochemical parameters to zero mean and unit variance: **a** monsoon and **b** non-monsoon

Table 4 Loadings of the significant principal components for water quality parameters in the Yamuna River. Italicized values represent variables with absolute loadings ≥ 0.3

	Monsoon		Non-monsoon	
	PC1	PC2	PC1	PC2
Temp.	0.1	0.1	0.2	−0.1
pH	*−0.3*	0.2	0.0	*−0.4*
DO	−0.1	*−0.4*	−0.2	*−0.3*
BOD	0.2	0.2	0.2	*0.3*
TDS	*0.3*	0.0	*0.3*	−0.1
EC	*0.3*	0.0	*0.3*	−0.1
TA	*0.3*	−0.2	*0.3*	−0.1
TH	*0.3*	−0.1	*0.3*	−0.1
Ca-H	0.2	*0.3*	*0.3*	0.1
Mg-H	0.2	*−0.3*	0.2	*−0.3*
NH_4^+	0.1	0.2	0.2	0.0
NO_2^-	0.2	0.2	0.1	*−0.3*
NO_3^-	0.0	*0.5*	0.1	*0.4*
PO_4^{3-}	0.2	*0.4*	0.2	*0.3*
SO_4^{2-}	*0.3*	0.0	*0.3*	−0.1
Cl^-	*0.3*	0.0	*0.3*	−0.1
F^-	0.2	0.0	*0.3*	0.0
Fe	−0.1	0.1	0.2	*0.3*
B	0.2	*−0.3*	0.2	0.1
Eigenvalue	9.2	4.2	11.1	4.8
Variance (%)	48.6	22.0	58.6	25.1
Cumulative (%)	48.6	70.6	58.6	83.7

EC, TA, TH, SO_4^{2-}, and Cl^-. This observation indicated that the river was strongly influenced by the components of salts and minerals. Salt ions in the river can result from (a) natural sources, e.g., atmospheric deposition, and weathering of soils, rocks, and minerals; or (b) anthropogenic sources such as the application of deicing chemicals, landfills, and agriculture (Gholizadeh et al. 2016). PC2, accounting for 22.0% of the total variance (Fig. 4a), had strong loadings on DO, Ca-H, Mg-H, NO_3^-, PO_4^{3-}, and B. This result indicated that PC2 was affected by organic and nutrient substances and biological activities. Moreover, natural mineralization, discharge of sewage, and/or using fertilizers in agriculture contributed to the variance retained by PC2.

PCs for non-monsoon

PC1 explained 58.6% of the total variance, and it showed strong and positive loadings on TDS, EC, TA, TH, Ca-H, SO_4^{2-}, Cl^-, and F^-. This finding demonstrated that PC1 could describe the amount of dissolved mineral salts originating from a common source. The results of PC2 (25.1%) depicted that BOD_5, NO_3^-, PO_4^{3-}, and Fe were located on the opposite direction compared with other water quality parameters (Fig. 4b), suggesting that these components could be originated from distinctive anthropogenic factors.

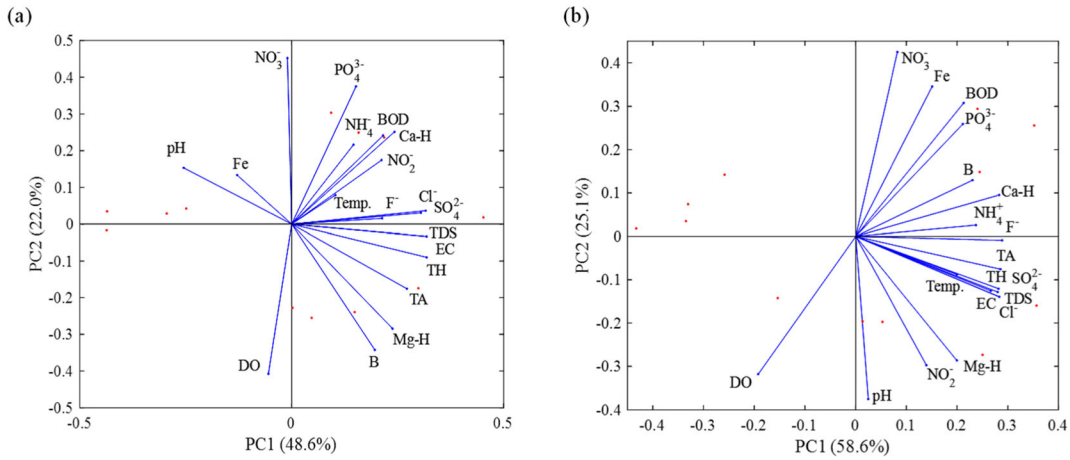

Fig. 4 Loading plot of water quality parameters in PCA: **a** monsoon and **b** non-monsoon

Cluster analysis for monsoon

The monitoring sites during the monsoon season were classified according to their water quality parameters into four groups (Fig. 5a):

Group M-1, consisting of Paonta, Kalanaur, Mawi, and Palla, was distributed on the negative side of PC1. This group having low TDS and organic matter was suitable for drinking, irrigation, and aquatic life (IS 1982).

Group M-2 had the locations of Delhi, Mohana, and Mathura, which were positioned on the upper part of PC2. These sites had high BOD_5 of 11.9 mg/L, NH_4^+ of 2.7 mg/L, and NO_3^- of 14.2 mg/L, but low DO of 4.9 mg/L, suggesting the contribution of an organic load from drains at the urban areas. Urban agglomerations such as Delhi and Mathura are significantly utilizing the Yamuna River for domestic water supply. Two major drains namely Najafgarh and Shahdara are continuously discharging a heavy load of contaminants into Yamuna at Delhi. Najafgarh Drain (28° 42′ 40″ N and 77° 13′ 55″ E) carries a flow of 20.7 m^3/s with an average BOD_5 load of 100 tons/day. The Shahdara Drain (28° 31′ 44″ N and 77° 16′ 57″ E) is the second main polluting source that receives partially treated effluents from industrial units in addition to discharges of minor drains carrying sewage and municipal wastes (Parmar and Bhardwaj 2015).

Group M-3, containing Etawah and Agra, was located at the right PC1 direction. This group was characterized by high dissolved salts expressed by an average TDS of 348 mg/L. The Yamuna water in these cities was

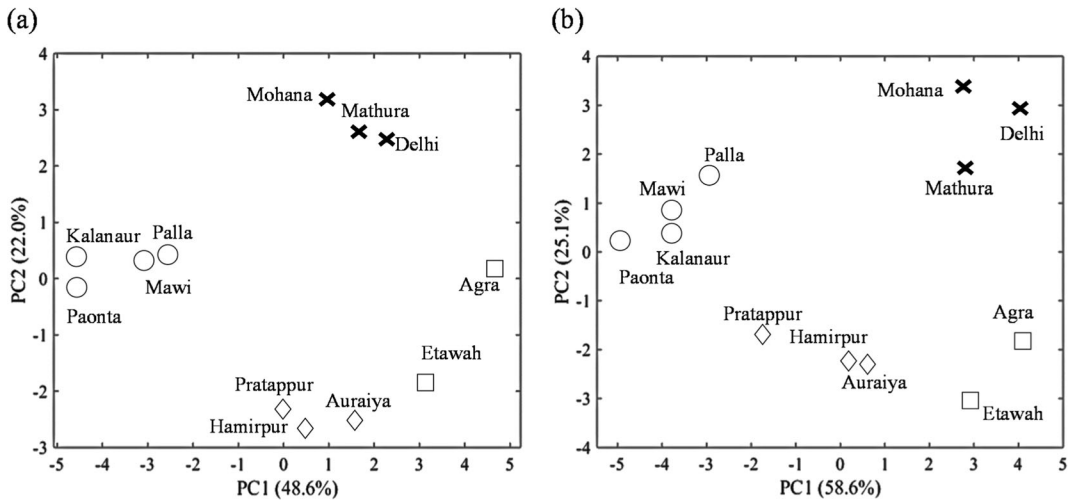

Fig. 5 Cluster analysis sampling sites: **a** monsoon and **b** non-monsoon

severely influenced by pollution coming from Delhi (located upstream). The cumulative wastewater discharge obtained from 22 major drains in Delhi was 49.57 m^3/sec during the year 2000, which decreased to 42.65 m^3/sec during 2005 (CPCB 2006). Moreover, this group was influenced by several pollution patterns resulting from the dumping of dead animals, agricultural runoffs, and cattle washing.

Group M-4 of Auraiya, Hamirpur, and Pratappur was located at the negative PC2. These sites contained high DO concentration with an average value of 7.3 mg/L. DO is considered the best index that describes the quality of aquatic environments because most organisms consume oxygen to metabolize food for producing energy and maintaining biological diversity. The increased values of DO could be due to the oxygenic photosynthesis of some aquatic plants and algae that utilize CO_2, water, and sunlight to produce oxygen (Ayers and Westcot 1994).

Cluster analysis for non-monsoon

The monitoring sites during the non-monsoon season were classified similar to those during monsoon (Fig. 5b):

Group NM-1 consisted of Paonta, Kalanaur, Mawi, and Palla. These sites are located at the upstream of the Yamuna River and described by low TA and TH values of 142 and 140 mg/L, respectively.

Group NM-2 had the sites of Delhi, Mohana, and Mathura and composed of high BOD_5, NH_4^+, and NO_3^- levels of 45.1, 3.1, and 14.4 mg/L, respectively. These sites are located at the midstream of the Yamuna River, and they were influenced by the discharge of domestic sewage from urbanized areas, runoff from fertilized agricultural lands, and erosion of soil containing nutrients.

Group NM-3 contained Agra and Etawah. This group is located at the midstream of the Yamuna River, and it was characterized by high TDS (907 mg/L), TA (294 mg/L), TH (291 mg/L), and EC (1512 μʊ/cm). The high Cl^- (209 mg/L) and SO_4^{2-} (100 mg/L) elements could be due to contamination with undetected and untreated pesticide residues that impose a toxic impact across the river. This pattern could be due to the discharge of three drains, namely Sarita Vihar, Sarita Vihar Bridge, and Kalkaji into the Agra canal.

Group NM-4 comprised Auraiya, Hamirpur, and Pratappur. These sites are located downstream of the Yamuna River, and they had low BOD_5 (1.78 mg/L), NH_4^+ (0.4 mg/L), NO_3^- (1.2 mg/L), and PO_4^{3-} (0.2 mg/L).

Water quality profile

The results of statistical analysis showed a high spatial variation of the physicochemical parameters along the entire Yamuna River stretch. The river was affected by pollution coming from untreated household sewage, industrial effluents, and fertilizers used in agriculture. The river was also influenced by multiple factors and environmental conditions including the properties of soil and landscape, human activities, water consumption, drainage systems, and geochemical characteristics of stream sediments. The river's longitudinal pollution gradient showed "excellent-to-good," "poor-to-marginal," and "good-to-excellent" WQIs at the upper, middle, and lower zones, respectively. The pollution gradient could be linked to the DO fluctuation (Supplementary file; Tables S2 and S3). The oxygen sag curve (Streeter and Phelps 1925) was plotted to describe the profile of DO concentrations along the Yamuna River starting from Palla (i.e., a site at which WQI initiated to decline). The rates of decomposition and reoxygenation in Eq. 2 were estimated following Parmar and Keshari (2012).

$$D = \frac{k_1 \times L_a}{k_2 - k_1}\left(e^{-k_1 \times t} - e^{-k_2 \times t}\right) + D_a \times e_{-k_2 \cdot t} \quad (2)$$

where D is saturation deficit (mg/L); k_1 and k_2 are deoxygenation and reaeration rates, respectively (1/day); L_a is the ultimate BOD (mg/L); D_a is initial oxygen deficit (mg/L); and t is travel time (day). DO was obtained from saturated DO minus D.

The DO longitudinal profile indicated that the river was healthy before Palla (Fig. 6). After this point, the DO concentration depleted due to organic decomposition (an indication of sewage disposal) with k_1 of 0.223 and 0.413 1 day for monsoon and non-monsoon, respectively. Similarly, Parmar and Keshari (2012) reported comparable k_1 values of 0.23–0.55 1 day for simulating the DO profile of the Yamuna River along the Delhi stretch. The decrease in water quality at the midstream of Yamuna could be due to the growing density of the human population on the river banks along with poor sanitation practices by inhabitants. Hence, this group should rely on groundwater for domestic purposes, and

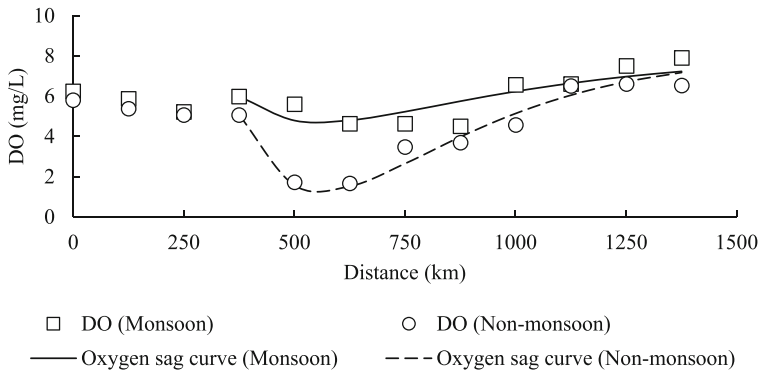

Fig. 6 DO profile along the Yamuna River and corresponding oxygen sag curve starting from Palla

the drinking water source should be subjected to conventional treatment and disinfection. Moreover, some sites in the middle zone were appropriate for the reproduction of wildlife and aquatic culture. The oxygen content of the river initiated to increase after Mohana due to the reoxygenation effect ($k_2 = 0.654$ 1 day for monsoon and 0.484 1 day for non-monsoon). The river assimilated the waste and recovered its initial environmental condition during the long journey. This finding revealed the positive impacts of dilution, self-purification, and depuration on the water quality at downstream stations. Accordingly, various locations at the upstream and downstream regions could be used for drinking purposes and outdoor bathing. The presented model offers several advantages because it can be applied worldwide to other water streams to estimate the critical time at which the minimum DO concentration occurs.

Comparison of the current study with the literature

The aforementioned findings complied with those reported in previous studies that have applied various statistical and modelling techniques for evaluating the river's pollution condition. For example, Paliwal et al. (2007) used a QUAL2E software to determine the pollution loads of the Yamuna River at the NCT of Delhi. Their study suggested (a) the treatment of the drainage water before joining the Yamuna River, and (b) the diversion of a considerable amount of water to the Agra canal for treatment. Their study also found that maintaining the flowrate in the river higher than 10 m^3/s could preserve its water quality. Sharma and Singh (2009) employed a STREAM II software to describe the pollution condition of the Yamuna River at the NCT of Delhi due to organic matter. Their study demonstrated that the main reason for the river's pollution could be linked to the high population density at Delhi and similar regions, causing the discharge of huge quantities of sewage into the river. Parmar and Bhardwaj (2015) found that the water quality of the Yamuna River was good at Hathnikund; deteriorated at Nizamuddin, Mazawali, and Agra; and recovered at Juhikha. Bhardwaj et al. (2017) found that the Yamuna River at the Delhi region was critically polluted with heavy metals due to effluents from several anthropogenic sources. Sharma et al. 2017a, b) used a QUAL2Kw software to simulate DO, BOD_5, nitrogenous compounds, and total coliform of the Yamuna River at certain districts of Palla, Okhla, Old Delhi Railway Bridge, and Nizamuddin. However, the current study is considered the primary effort that described the pollution status of the whole stretch of the Yamuna River basin from Paonta to Pratappur.

Conclusions

This study demonstrated the effectiveness of multivariate statistical techniques for evaluating the water quality of the Yamuna River during the monsoon and non-monsoon seasons. The findings of PCA depicted that the river was primarily influenced by dissolved salts and minerals followed by organic and nutrient substances. The spatial variation of the physicochemical parameters indicated that the river could be classified into four groups (from upstream to downstream): (a) group 1, consisting of Paonta, Kalanaur, Mawi, and Palla, was suitable for drinking, irrigation, and aquatic life, (b) group 2, containing Delhi, Mohana, and Mathura, was negatively influenced by high organic loads coming from several drains such as Najafgarh and Shahdara, (c) group 3, containing Agra and Etawah, was

negatively impacted by high pollution resulting from domestic sewage, agricultural runoffs, and untreated pesticide, and (d) group 4, comprising Auraiya, Hamirpur, and Pratappur, had improved water quality conditions. Hence, the WQIs of the upper zone (from Paonta to Palla), midstream (from Delhi to Etawha), and the lower region (from Auraiya to Pratappur) were "excellent-to-good," "poor-to-marginal," and "good-to-excellent," respectively. The longitudinal pollution gradients complied with the oxygen sag curve, indicating that the river assimilated the waste load and recovered its initial characteristics in the downstream stations. The proposed statistical techniques could be usefully applied to analyze and interpret the water quality data and to estimate the critical time at which the minimum DO concentration occurs. Moreover, the information attained through this study can be employed to enhance the water quality management of water bodies and to identify the major pollution sources in future investigations.

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